

A Dataset Statistics

Table 1 summarizes the benchmark datasets used in our experiments, including the number of variables or channels (#Vars.), the number of observations (#Obs.), the sampling frequency (Freq.), and the application domain. For the Wiki dataset, we randomly sample 2,000 variables from the full dataset, which contains more than 100,000 time series. We provide the processed datasets used in our experiments at <https://figshare.com/s/7617530bce306584fe95?file=62576929>.

Table 1. Summary of benchmark datasets. ECL denotes Electricity.

	ECG	COVID-19	Solar	ECL	Traffic	METR-LA	PEMS-BAY	Wiki
#Vars.	140	55	593	370	963	207	325	2 000
#Obs.	5 000	335	3 650	140 211	10 560	34 272	52 116	803
Freq.	–	1 day	1 hr	15 min	1 hr	5 min	5 min	1 day
Domain	Bio.	Epid.	Energy	Energy	Traffic	Traffic	Traffic	Web

B Evaluation Metrics

We evaluate all models using two complementary metrics: the Coefficient of Determination (R^2) and Mean Absolute Error (MAE). Following the notation introduced in Section 3.1, let N denote the number of variables, O the forecasting horizon, and $\hat{\mathbf{Y}} = F_\theta(\mathbf{X}_{\text{in}}) \in \mathbb{R}^{B \times O \times N}$ the model predictions. We write $Y_{o,n}^b$ and $\hat{Y}_{o,n}^b$ for the ground-truth and predicted value of variable n at horizon step o for sample b , respectively.

Coefficient of Determination (R^2). R^2 measures the proportion of variance in the ground-truth targets explained by the model. Higher values indicate better fit, with a maximum of 1.

$$R^2 = 1 - \frac{\sum_{b=1}^B \sum_{o=1}^O \sum_{n=1}^N (Y_{o,n}^b - \hat{Y}_{o,n}^b)^2}{\sum_{b=1}^B \sum_{o=1}^O \sum_{n=1}^N (Y_{o,n}^b - \bar{Y})^2}, \quad (1)$$

where

$$\bar{Y} = \frac{1}{BON} \sum_{b=1}^B \sum_{o=1}^O \sum_{n=1}^N Y_{o,n}^b$$

is the global mean of the ground-truth targets over the test set.

Mean Absolute Error (MAE). MAE measures the average absolute difference between the predicted and ground-truth values. Lower values indicate better performance.

$$\text{MAE} = \frac{1}{BON} \sum_{b=1}^B \sum_{o=1}^O \sum_{n=1}^N \left| Y_{o,n}^b - \hat{Y}_{o,n}^b \right|. \quad (2)$$

C Supplementary Results

This section provides additional experimental results. Section C.1 presents the main-paper results with standard deviations, and Section C.2 reports results across forecast horizons $\{6, 12, 24, 48\}$, also with standard deviations.

C.1 Main Results with Standard Deviations

Tables 2 and 3 report the results for the setting with input length 12 and forecast horizon 12. In addition to the main-paper results, they also include the corresponding mean and standard deviation over 5 random seeds.

C.2 Results Across Forecast Horizons

Tables 4–7 report forecasting performance across forecast horizons $\{6, 12, 24, 48\}$. Results are averaged over 3 random seeds and reported as mean $\pm$ standard deviation, except for the horizon-12 setting in the main paper, which is averaged over 5 random seeds. Avg. denotes the mean score over the four forecast horizons. The compared methods include the top four baseline models for each dataset and our two proposed models.

Table 2. Forecasting results on ECG, COVID-19 (COVID), Solar, and Electricity (ECL), averaged over 5 runs. Values are reported as mean $\pm$ standard deviation. **Bold** and underline indicate the best and second-best results in each column, respectively.

Models	Metric	ECG	COVID	Solar	ECL
FourierGNN	$R^2\uparrow$	<u>.707</u> $\pm$.0002	.950 $\pm$.0056	.742 $\pm$.0010	.912 $\pm$.0013
	MAE $\downarrow$	.329 $\pm$.0001	31.50 $\pm$ 1.575	9.03 $\pm$.0193	97.13 $\pm$.474
Spike-GRU	$R^2\uparrow$	.706 $\pm$.0002	.845 $\pm$.0036	.487 $\pm$.0045	.898 $\pm$.0016
	MAE $\downarrow$	.331 $\pm$.0002	55.35 $\pm$.839	14.34 $\pm$.1003	107.09 $\pm$ 1.04
iSpikformer	$R^2\uparrow$	<u>.707</u> $\pm$.0002	.874 $\pm$.0237	.593 $\pm$.0755	.829 $\pm$.0195
	MAE $\downarrow$	.330 $\pm$.0006	49.79 $\pm$ 5.166	12.27 $\pm$ 1.6421	139.73 $\pm$ 9.25
SpikeRNN w/ CPG	$R^2\uparrow$	<u>.707</u> $\pm$.0000	.836 $\pm$.0031	.680 $\pm$.0004	.916 $\pm$.0037
	MAE $\downarrow$	.330 $\pm$.0004	57.87 $\pm$.458	10.37 $\pm$.0862	96.45 $\pm$ 1.21
SpikeTCN w/ CPG	$R^2\uparrow$	<u>.707</u> $\pm$.0004	.849 $\pm$.0068	.706 $\pm$.0061	.910 $\pm$.0059
	MAE $\downarrow$	.330 $\pm$.0006	55.20 $\pm$ 1.964	9.84 $\pm$.0984	96.17 $\pm$ 3.26
Spikformer w/ CPG	$R^2\uparrow$	.706 $\pm$.0008	.867 $\pm$.0022	.699 $\pm$.0033	<u>.930</u> $\pm$.0057
	MAE $\downarrow$	.330 $\pm$.0006	48.71 $\pm$.489	10.03 $\pm$.1568	84.65 $\pm$.429
TS-GRU	$R^2\uparrow$	<u>.707</u> $\pm$.0000	.842 $\pm$.0061	.590 $\pm$.0212	.902 $\pm$.0021
	MAE $\downarrow$	.329 $\pm$.0001	56.81 $\pm$.924	12.13 $\pm$.3755	103.36 $\pm$ 1.10
TS-TCN	$R^2\uparrow$	<u>.707</u> $\pm$.0000	.836 $\pm$.0021	.707 $\pm$.0071	.893 $\pm$.0387
	MAE $\downarrow$	.330 $\pm$.0003	57.89 $\pm$.436	9.80 $\pm$.1001	104.51 $\pm$ 21.02
TS-Former	$R^2\uparrow$	.704 $\pm$.0012	.835 $\pm$.0002	.679 $\pm$.0231	.925 $\pm$.0086
	MAE $\downarrow$	.330 $\pm$.0009	58.02 $\pm$.017	10.50 $\pm$.4997	<u>84.78</u> $\pm$.615
SpikF	$R^2\uparrow$	.708 $\pm$.0001	.961 $\pm$.0062	.712 $\pm$.0012	.910 $\pm$.0011
	MAE $\downarrow$	<u>.327</u> $\pm$.0001	28.16 $\pm$ 1.802	9.61 $\pm$.0324	99.99 $\pm$.115
SpikF-GO	$R^2\uparrow$	.708 $\pm$.0002	.967 $\pm$.0042	<u>.740</u> $\pm$.0048	.927 $\pm$.0068
	MAE $\downarrow$	<u>.327</u> $\pm$.0001	25.15 $\pm$ 1.654	<u>9.18</u> $\pm$.0648	91.15 $\pm$ 1.29
SpikF-GO w/ CPG	$R^2\uparrow$	.708 $\pm$.0005	<u>.965</u> $\pm$.0024	.742 $\pm$.0051	.942 $\pm$.0028
	MAE $\downarrow$	.326 $\pm$.0003	<u>26.09</u> $\pm$.703	9.03 $\pm$.1060	89.14 $\pm$ 1.08

Table 3. Forecasting results on METR-LA (METR), Traffic, PEMS-BAY (PEMS), and Wiki, averaged over 5 runs. Values are reported as mean $\pm$ standard deviation. **Bold** and underline indicate the best and second-best results in each column, respectively.

Models	Metric	METR	Traffic	PEMS	Wiki
FourierGNN	$R^2\uparrow$	.766 $\pm$.0003	.632 $\pm$.0032	.733 $\pm$.0032	.473 $\pm$.0063
	MAE $\downarrow$	.071 $\pm$.0001	.013 $\pm$.0001	2.16 $\pm$.0023	185.64 $\pm$.81
Spike-GRU	$R^2\uparrow$	.691 $\pm$.0022	.491 $\pm$.0452	.573 $\pm$.0033	.458 $\pm$.0017
	MAE $\downarrow$	.086 $\pm$.0004	.020 $\pm$.0004	2.91 $\pm$.0201	201.08 $\pm$ 1.18
iSpikformer	$R^2\uparrow$	.748 $\pm$.0175	.483 $\pm$.0124	.739 $\pm$.0059	.469 $\pm$.0058
	MAE $\downarrow$	.075 $\pm$.0028	.018 $\pm$.0004	2.25 $\pm$.0298	195.26 $\pm$ 1.92
SpikeRNN w/ CPG	$R^2\uparrow$	.728 $\pm$.0088	.553 $\pm$.0376	.732 $\pm$.0020	.458 $\pm$.0010
	MAE $\downarrow$	.079 $\pm$.0016	.017 $\pm$.0009	2.30 $\pm$.0079	202.26 $\pm$ 1.26
SpikeTCN w/ CPG	$R^2\uparrow$	.759 $\pm$.0012	.688 $\pm$.0166	.747 $\pm$.0008	.447 $\pm$.0076
	MAE $\downarrow$	.073 $\pm$.0002	.014 $\pm$.0008	2.25 $\pm$.0017	218.76 $\pm$ 5.52
Spikformer w/ CPG	$R^2\uparrow$	.747 $\pm$.0030	.728 $\pm$.0109	.721 $\pm$.0046	.467 $\pm$.0032
	MAE $\downarrow$	.076 $\pm$.0006	<u>.012</u> $\pm$.0002	2.28 $\pm$.0141	202.63 $\pm$ 1.43
TS-GRU	$R^2\uparrow$	.758 $\pm$.0006	.437 $\pm$.0065	.735 $\pm$.0002	.451 $\pm$.0029
	MAE $\downarrow$	.074 $\pm$.0002	.019 $\pm$.0002	2.25 $\pm$.0020	197.42 $\pm$.411
TS-TCN	$R^2\uparrow$	.692 $\pm$.0000	.689 $\pm$.0061	.743 $\pm$.0041	.459 $\pm$.0002
	MAE $\downarrow$	.086 $\pm$.0000	.014 $\pm$.0002	2.27 $\pm$.0076	204.46 $\pm$.900
TS-Former	$R^2\uparrow$	.747 $\pm$.0028	.744 $\pm$.0098	.723 $\pm$.0051	.465 $\pm$.0020
	MAE $\downarrow$	.076 $\pm$.0007	.011 $\pm$.0003	2.29 $\pm$.0199	206.41 $\pm$ 1.35
SpikF	$R^2\uparrow$	.766 $\pm$.0003	.541 $\pm$.0019	.740 $\pm$.0008	.477 $\pm$.0047
	MAE $\downarrow$	<u>.072</u> $\pm$.0001	.016 $\pm$.0001	2.21 $\pm$.0014	<u>186.58</u> $\pm$.700
SpikF-GO	$R^2\uparrow$	.762 $\pm$.0160	.651 $\pm$.0032	<u>.762</u> $\pm$.0017	.480 $\pm$.0097
	MAE $\downarrow$	.077 $\pm$.0037	.013 $\pm$.0000	<u>2.17</u> $\pm$.0081	189.22 $\pm$ 3.28
SpikF-GO w/ CPG	$R^2\uparrow$	.769 $\pm$.0010	.669 $\pm$.0085	.766 $\pm$.0018	.466 $\pm$.0154
	MAE $\downarrow$	<u>.072</u> $\pm$.0004	<u>.012</u> $\pm$.0002	2.16 $\pm$.0116	191.21 $\pm$ 3.56

Table 4. COVID-19 results across various forecast horizons. Avg. averages the mean scores over horizons {6, 12, 24, 48}; standard deviations are shown only for individual horizons. **Bold**/underline denote best/second-best based on the unrounded mean values.

Models	Metric	6	12	24	48	Avg.
FourierGNN	$R^2\uparrow$	.990 $\pm$.002	.950 $\pm$.006	.764 $\pm$.006	.408 $\pm$.002	.778
	MAE $\downarrow$	16.56 $\pm$ 1.3	31.50 $\pm$ 1.6	62.23 $\pm$ 1.1	103.55 $\pm$.4	53.5
iSpikformer	$R^2\uparrow$	.947 $\pm$.017	.874 $\pm$.024	.689 $\pm$.021	.393 $\pm$.004	.726
	MAE $\downarrow$	34.04 $\pm$ 5.9	49.79 $\pm$ 5.2	73.47 $\pm$ 3.0	106.38 $\pm$.6	65.9
TS-GRU	$R^2\uparrow$	.920 $\pm$.007	.842 $\pm$.006	.651 $\pm$.002	.382 $\pm$.001	.699
	MAE $\downarrow$	42.72 $\pm$ 1.6	56.81 $\pm$.92	79.24 $\pm$.26	108.65 $\pm$.2	71.9
SpikF	$R^2\uparrow$	.991 $\pm$.004	.961 $\pm$.006	.782 $\pm$.012	.408 $\pm$.002	.786
	MAE $\downarrow$	15.68 $\pm$ 2.3	28.16 $\pm$ 1.8	59.83 $\pm$ 1.6	103.58 $\pm$.3	51.8
SpikF-GO	$R^2\uparrow$	.991 $\pm$.004	.967 $\pm$.004	.794 $\pm$.018	<u>.410</u> $\pm$.002	<u>.791</u>
	MAE $\downarrow$	<u>15.58</u> $\pm$ 2.6	25.15 $\pm$ 1.7	56.45 $\pm$ 3.0	<u>103.18</u> $\pm$.5	50.1
SpikF-GO w/ CPG	$R^2\uparrow$	.991 $\pm$.003	.965 $\pm$.002	.788 $\pm$.021	.420 $\pm$.006	.791
	MAE $\downarrow$	15.51 $\pm$ 1.6	26.09 $\pm$.70	<u>58.44</u> $\pm$ 3.4	101.29 $\pm$ 1.	<u>50.3</u>

Table 5. PEMS-BAY results across various forecast horizons. Avg. averages the mean scores over horizons {6, 12, 24, 48}; standard deviations are shown only for individual horizons. **Bold/underline** denote best/second-best based on the unrounded mean values.

Models	Metric	6	12	24	48	Avg.
FourierGNN	$R^2\uparrow$	.847 $\pm$.001	.733 $\pm$.003	.508 $\pm$.001	.123 $\pm$.008	.553
	MAE $\downarrow$	1.62 $\pm$.002	2.16 $\pm$.002	3.04 $\pm$.002	4.39 $\pm$.010	2.80
SpikeTCN	$R^2\uparrow$	.857 $\pm$.000	.747 $\pm$.001	.539 $\pm$.001	.078 $\pm$.012	.555
	MAE $\downarrow$	1.72 $\pm$.000	2.25 $\pm$.002	3.09 $\pm$.008	4.48 $\pm$.013	2.89
TS-TCN	$R^2\uparrow$	.854 $\pm$.003	.743 $\pm$.004	.520 $\pm$.013	.189 $\pm$.000	.577
	MAE $\downarrow$	1.74 $\pm$.012	2.27 $\pm$.008	3.13 $\pm$.014	4.46 $\pm$.000	2.90
SpikF	$R^2\uparrow$	.862 $\pm$.001	.740 $\pm$.001	.504 $\pm$.000	.120 $\pm$.000	.557
	MAE $\downarrow$	<u>1.66</u> $\pm$.001	2.21 $\pm$.001	3.10 $\pm$.001	4.41 $\pm$.001	2.85
SpikF-GO	$R^2\uparrow$	<u>.870</u> $\pm$.001	<u>.762</u> $\pm$.002	<u>.545</u> $\pm$.004	<u>.209</u> $\pm$.005	<u>.597</u>
	MAE $\downarrow$	1.62 $\pm$.000	<u>2.17</u> $\pm$.008	<u>3.03</u> $\pm$.002	4.33 $\pm$.010	<u>2.79</u>
SpikF-GO w/ CPG	$R^2\uparrow$	.873 $\pm$.000	.766 $\pm$.002	.569 $\pm$.000	.225 $\pm$.006	.608
	MAE $\downarrow$	1.62 $\pm$.003	2.16 $\pm$.012	3.02 $\pm$.023	<u>4.34</u> $\pm$.028	2.79

Table 6. Solar results across various forecast horizons. Avg. averages the mean scores over horizons {6, 12, 24, 48}; standard deviations are shown only for individual horizons. **Bold/underline** denote best/second-best based on the unrounded mean values.

Models	Metric	6	12	24	48	Avg.
FourierGNN	$R^2\uparrow$	.767 $\pm$.001	.742 $\pm$.001	<u>.703</u> $\pm$.001	.670 $\pm$.001	.721
	MAE $\downarrow$	8.56 $\pm$.008	9.03 $\pm$.019	<u>9.80</u> $\pm$.020	<u>10.47</u> $\pm$.02	<u>9.47</u>
SpikeTCN	$R^2\uparrow$	.743 $\pm$.007	.706 $\pm$.006	.656 $\pm$.005	.620 $\pm$.007	.681
	MAE $\downarrow$	9.26 $\pm$.148	9.84 $\pm$.098	10.74 $\pm$.12	11.29 $\pm$.09	10.28
TS-TCN	$R^2\uparrow$	.740 $\pm$.004	.707 $\pm$.007	.658 $\pm$.004	.621 $\pm$.006	.682
	MAE $\downarrow$	9.28 $\pm$.089	9.80 $\pm$.100	10.67 $\pm$.11	11.30 $\pm$.06	10.26
SpikF	$R^2\uparrow$	.735 $\pm$.000	.712 $\pm$.001	.667 $\pm$.002	.633 $\pm$.001	.687
	MAE $\downarrow$	9.20 $\pm$.022	9.61 $\pm$.032	10.43 $\pm$.045	11.14 $\pm$.024	10.10
SpikF-GO	$R^2\uparrow$	<u>.765</u> $\pm$.004	<u>.740</u> $\pm$.005	<u>.702</u> $\pm$.008	<u>.665</u> $\pm$.014	.718
	MAE $\downarrow$	8.61 $\pm$.068	9.18 $\pm$.065	9.87 $\pm$.098	10.51 $\pm$.164	9.54
SpikF-GO w/ CPG	$R^2\uparrow$	.767 $\pm$.011	.742 $\pm$.005	.707 $\pm$.016	<u>.665</u> $\pm$.028	<u>.720</u>
	MAE $\downarrow$	<u>8.60</u> $\pm$.144	9.03 $\pm$.106	9.66 $\pm$.203	10.40 $\pm$.37	9.42

Table 7. Electricity results across various forecast horizons. Avg. averages the mean scores over horizons {6, 12, 24, 48}; standard deviations are shown only for individual horizons. **Bold/underline** denote best/second-best based on the unrounded mean values.

Models	Metric	6	12	24	48	Avg.
FourierGNN	$R^2\uparrow$	.959 $\pm$.000	.912 $\pm$.001	.799 $\pm$.000	.598 $\pm$.002	.817
	MAE $\downarrow$	65.94 $\pm$.20	97.13 $\pm$.47	157.46 $\pm$.1	241.78 $\pm$.0	141
Spikformer w/ CPG	$R^2\uparrow$	.963 $\pm$.003	<u>.930$\pm$.006</u>	<u>.858$\pm$.006</u>	.761$\pm$.011	<u>.878</u>
	MAE $\downarrow$	65.59 $\pm$ 1.1	84.65$\pm$.43	118.99$\pm$2.	168.92$\pm$.8	110
TS-Former	$R^2\uparrow$	.961 $\pm$.005	.925 $\pm$.009	.835 $\pm$.006	.738 $\pm$.012	.865
	MAE $\downarrow$	64.91 $\pm$ 1.1	<u>84.78$\pm$.62</u>	<u>121.25$\pm$.4</u>	<u>170.19$\pm$3.</u>	<u>110</u>
SpikF	$R^2\uparrow$	.958 $\pm$.000	.910 $\pm$.001	.795 $\pm$.002	.596 $\pm$.000	.815
	MAE $\downarrow$	69.35 $\pm$.08	99.99 $\pm$.12	157.91 $\pm$.0	238.06 $\pm$.1	141
SpikF-GO	$R^2\uparrow$	.967 $\pm$.001	.927 $\pm$.007	.834 $\pm$.005	.718 $\pm$.000	.862
	MAE $\downarrow$	63.25$\pm$.19	91.15 $\pm$ 1.3	144.60 $\pm$ 2.	213.32 $\pm$.9	128
SpikF-GO w/ CPG	$R^2\uparrow$	.971$\pm$.001	.942$\pm$.003	.859$\pm$.003	<u>.745$\pm$.001</u>	.879
	MAE $\downarrow$	<u>64.29$\pm$.28</u>	89.14 $\pm$ 1.1	142.01 $\pm$.1	211.64 $\pm$ 1.	127